Paper Summary: Reimbursement Policy and Drug Shortages

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In one sentence: This paper builds a game-theoretic drug-supply-chain model (manufacturers – GPOs – providers – Medicare) to characterize how Medicare's ASP (average sales price) reimbursement policy—a two-tier "wholesale-price → reimbursement-price" structure—shapes wholesale-pricing decisions and drug shortages. It identifies two opposing forces, the **free-ride effect** and the **aligning effect**, and (counterintuitively) argues that ASP actually possesses **resistance** to shortages for the drugs that currently experience them, implying ASP is unlikely to be the major cause.

1. Paper Type

Theory (core) | Data-calibrated numerical study (supplement)

A **theory paper** at its core—it derives equilibrium existence/uniqueness (Theorem 1) and conducts qualitative comparative-statics analysis (Theorems 2–3) with managerial interpretation. It is complemented by a **data-driven numerical study** (Section 5) that uses real data and machine learning to estimate a key parameter (n, the number of GPOs) and to run numerical experiments.

2. Motivation

- **Real-world pain point:** U.S. drug shortages have persisted for two decades—active shortages tripled from 2005 to 2010, hit 456 in 2012, stayed above 300 since 2018, and reached 323 in Q1 2024 (the highest in nearly a decade). They cause treatment delays, medication-safety issues, and even patient deaths, plus ~$360M annual labor cost and 5%–20% higher procurement budgets.
- **Long-standing policy debate:** In 2005 Medicare switched from the AWP (average wholesale price) scheme—reimbursing at a fixed published list price—to the ASP (average sales price) scheme, which reimburses at 106% of the actual average wholesale price over the prior two quarters. One camp blames ASP for depressing prices and discouraging manufacturers (worsening shortages); the other argues ASP does not directly regulate prices (it only requires reimbursement 6% above the average wholesale price), preserving pricing flexibility. The debate lacks rigorous resolution.
- **Research gap:** The FDA (2020) states that "quantitative data about the relationship between reimbursement policy and shortages is scarce," making empirical work very hard. Almost no

research models **wholesale-pricing decisions** under the reimbursement policy, and the unique role of GPOs (group purchasing organizations)—a key but poorly understood party, now under FTC/HHS investigation—has been largely overlooked.

3. Research Question

(1) Capture the essential elements and tradeoffs in drug **wholesale-pricing decisions** under the reimbursement policy;
(2) Investigate the impact of the **reimbursement policy** and the **unique role of GPOs** on drug shortages via a supply-chain model.
Central question: does (and how does) the ASP reimbursement policy cause/worsen drug shortages, and how do GPOs' pricing decisions mediate this?

4. Model & Theory

4.1 System structure — the parties

The system is a drug supply chain with **four parties**: (i) drug **manufacturers** (suppliers); (ii) n symmetric **GPOs**, which represent their provider members and set wholesale prices with manufacturers; (iii) **healthcare providers** (hospitals serving patients); and (iv) the **government (Medicare)**, which reimburses drug sales. Figure 1 (reproduced from the paper) summarizes the structure and parameters.

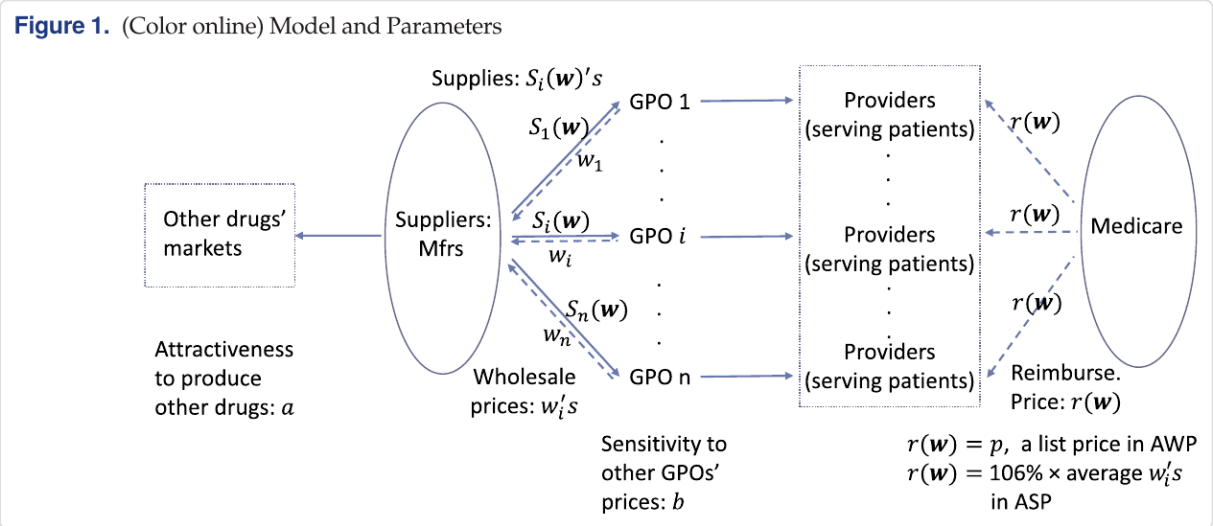


Figure 1 (reproduced from Zhao, Jia & Zhao 2025): the four-party drug supply chain and key parameters.

4.2 Dynamics of the system (sequence of actions)

The paper models this as a **single-period** game (the period corresponds to a 2–3-year contract duration, over which wholesale prices are fixed). The actions unfold in the following order:

- 1 **GPOs set wholesale prices.** Each GPO i pools demand from its members and negotiates/sets a wholesale price w_i with manufacturers (GPOs are the effective price setters

for these generic drugs).

- ② **Manufacturers decide supply allocation.** Given the vector of wholesale prices $\mathbf{w}$, manufacturers determine how much supply $S_i(\mathbf{w})$ to allocate to each GPO—more to GPOs offering higher wholesale prices, less when other drugs are more attractive to produce.
- ③ **Providers meet patient demand.** Providers procure through their GPOs to serve a (price-independent, deterministic) demand d; realized sales equal the minimum of supply and demand, $S_i(\mathbf{w}) \wedge d$.
- ④ **Medicare reimburses.** The government pays the reimbursement price $r(\mathbf{w})$: under AWP, $r = p$ (a fixed list price); under ASP, $r = \theta \cdot \bar{w}$ (106% of the average wholesale price).
- ⑤ **Shortage is determined** by whether supply is systematically insufficient to meet demand.

4.3 Key variables

Symbol	Meaning	Notes
w_i / $\mathbf{w}$	Wholesale price set by GPO i / vector of all GPOs' prices	GPO's decision variable
$r(\mathbf{w})$	Reimbursement price	AWP: $r = p$ (fixed list price); ASP: $r = \theta \cdot$(avg wholesale price $\bar{w}$)
θ	Reimbursement markup percentage	Currently 106%; treated as a variable policy parameter
d	Demand per GPO	Price-independent, deterministic (medically necessary, low elasticity)
n	Number of (symmetric) major GPOs in a drug's market	Key driver of the free-ride effect
α	GPO's "price consciousness" (weight on members' profit, $0 \leqslant \alpha \leqslant 1$)	Larger α = more focus on pushing wholesale price down
γ	Commission percentage (contract administration fee)	Capped at 3% by the 1987 safe-harbor provision
$S_i(\mathbf{w})$	Supply allocated to GPO i by manufacturers	Eq. (3): allocation term × random factor
a	Attractiveness of producing other drugs (between-drug competition)	Supply-difficulty parameter; larger $a \rightarrow$ less supply for the focal drug
b	Manufacturers' sensitivity to other GPOs' prices (between-GPO competition), $0 \leqslant b < 1$	Supply-difficulty parameter
ε_i	Nonnegative random factor with mean μ_i	Captures high supply randomness (multiplicative)

4.4 GPO objective function (Eqs. 1–2)

$$\pi_i(\mathbf{w}) = \alpha \cdot [r(\mathbf{w}) - w_i] \cdot E[S_i \wedge d] \; + \; (1-\alpha) \cdot \gamma \cdot w_i \cdot E[S_i \wedge d]$$

The payoff is a weighted sum of two parts:

- **Members' profit** (weight α): $[r(\mathbf{w}) - w_i] \cdot E[S_i \wedge d]$ — the per-unit margin (reimbursement price minus wholesale price) times sales volume. Higher wholesale prices hurt this term.
- **GPO's own profit** (weight $1-\alpha$): $\gamma \cdot w_i \cdot E[S_i \wedge d]$ — the commission (contract administration fee) γ on member purchases, which is a percentage of the wholesale price. Higher wholesale prices help this term.
- **Sales volume** $E[S_i \wedge d]$: the minimum of supply and demand—GPOs cannot sell more than what is supplied or demanded.

Maximizing this is equivalent (Eq. 2) to $\max_{w_i} \{\alpha r(\mathbf{w}) - [\alpha-(1-\alpha)\gamma]w_i\} \cdot E[S_i \wedge d]$, where the bracketed term is the GPO's "unit payoff."

Supply-allocation model (Eq. 3):

$$S_i(\mathbf{w}) = (w_i - b \cdot \bar{w}_{-i} - a) \times \varepsilon_i$$

Supply rises with the GPO's own wholesale price, falls with other GPOs' prices (via b) and with the attractiveness of producing other drugs (a); the multiplicative random factor ε_i keeps supply well-formed.

4.5 Two GPO types

- **Member-centric:** the unit payoff decreases in w_i, so the GPO must trade off members' profit —this is the analytical focus and the dominant type in practice.
- **Self-centered:** the unit payoff increases in w_i, so it is optimal to push the wholesale price to its upper bound $\hat{w}$. Occurs only when α is very small ($\alpha \le \gamma/(1+\gamma)$) or n is very small (e.g., $n=1$ monopoly).

4.6 Shortage measures

- **Long-term shortage occurrence:** indicator $1\{ (w_i - b \cdot \bar{w}_{-i} - a) \cdot \mu_i \le \kappa d \}$ — a systematic/long-term shortage occurs when mean supply falls considerably below demand (below a fraction κ of it).
- **Long-term shortage magnitude:** $[1 - (w_i - b \cdot \bar{w}_{-i} - a) \cdot \mu_i / d]^+$. The two measures are positively correlated, so directional results use "shortage" generically.

4.7 Core results

Theorem 1 (Equilibrium): For member-centric GPOs, a **unique symmetric interior equilibrium** wholesale price exists under both ASP and AWP (each characterized by an implicit equation). For self-centered GPOs, a unique dominant-strategy equilibrium $w_i = \hat{w}$ exists.

Theorem 2 (Policy impact, member-centric):
(i) The equilibrium wholesale price **increases** in a and b under both ASP and AWP; however, **shortage increases** in a, b under AWP but **decreases** in a, b under ASP. ← the

most counterintuitive and central result.
(ii) Under AWP, wholesale price and shortage are unaffected by n; under ASP, a larger n → lower wholesale price and higher shortage.

Two effects unique to ASP (and their intuition):
· **Free-ride effect:** since the reimbursement price is a markup on the *average* wholesale price, each GPO wants to set a *low* wholesale price while hoping others keep theirs high (to sustain a high average, hence a high reimbursement price). In equilibrium everyone under-prices → wholesale price ↓ → **shortage ↑ (proshortage)**.
· **Aligning effect:** if all GPOs raise prices together, all share a higher average and reimbursement price → wholesale price ↑ → **shortage ↓ (antishortage)**. But alignment is not the natural equilibrium outcome; it needs an external "stimulus"—**supply difficulty (a, b) is that stimulus**.

Why does higher supply difficulty reduce shortage under ASP (unpacking Thm 2i)?
Supply difficulty acts through two channels: a *direct* channel (harder to secure supply → more shortage) and an *indirect* channel (it triggers the aligning effect → GPOs raise wholesale prices → higher reimbursement margin → incentive to secure more supply → less shortage). Which channel wins differs by policy and by n:

- **Under ASP:** the **indirect (aligning) channel dominates**, so shortage falls in a, b—the source of ASP's "resistance."
- **Under AWP:** the reimbursement price is fixed, so there is no aligning loop and the direct channel dominates (shortage rises).
- **For n:** a larger n means each GPO has a smaller weight in the average price, so it treats reimbursement as more exogenous, strengthening the free-ride incentive; at $n=1$ (monopoly) free-riding is impossible.

Theorem 3 (Industry/policy parameters):
(i) Under both ASP and AWP, higher price consciousness α → lower equilibrium wholesale price and higher shortage (over-emphasizing price cuts backfires).
(ii) Under ASP, higher reimbursement markup θ → higher equilibrium wholesale price and lower shortage.

Corollary 1 (self-centered): $w_i=\hat{w}$ for all i and no shortage occurs; wholesale price and shortage are unaffected by drug-specific parameters (a monopoly GPO is always self-centered because it controls the reimbursement price).

5. Numerical Results (Section 5)

- **Data challenge:** pharmaceutical data are opaque (Social Security Act § 1927(b)(3)(D) restricts GPO data). The authors integrate Medicare Part B physician-usage data (2012–2016, HCPCS codes) with AHA hospital–GPO affiliation data, using **machine learning (SVM, chosen over**

random forest and neural network) to assign the 90% of physicians without GPO labels, yielding usage volume per drug per GPO for 452 drugs.

- **Estimating n:** on average about **3 GPOs (precisely 2.5) dominate 80% of each drug's market**—consistent with "top 5 GPOs hold 85%–90%" and the 80/20 rule; >30% of drugs have only one major GPO, and n rarely exceeds 5 → the **free-ride effect is weak in reality** → ASP resists shortages for these drugs → **ASP is unlikely to be the major cause**. Estimating n only for shortage drugs still gives ~3; results also hold without ML (robust).
- **a and b cannot be estimated** (data requirements are prohibitive), so they are swept over wide ranges numerically.
- **Figure 3:** the ASP shortage region expands with n (stronger free-riding); the AWP region is unaffected by n and points the opposite way (high a/b = shortage).
- **Figure 4:** raising θ shrinks the shortage region, with diminishing returns at high θ (must be weighed against higher Medicare spending).
- **Figure 5:** lowering α shrinks the shortage region, and the effect is larger under ASP than AWP. $\theta=106\%$, $\alpha=1$ is the worst case for ASP's resistance.
- **Figure 6 (shortage magnitude — important caveat):** *when* a shortage does occur, its magnitude under ASP can be **larger** for drugs with **moderate supply difficulty** (e.g., nervous-system, cardiovascular drugs).

  > **Intuition for this caveat:** resistance is strongest at the extremes, but weakest in the middle:
  >
  > - **At the extremes of supply difficulty:** either very low difficulty (little shortage anyway) or very high difficulty (the strong aligning effect lifts prices enough to protect supply)—ASP's resistance holds.
  > - **In the middle range:** the aligning effect is only partially triggered—not strong enough to fully offset the direct supply squeeze, yet the drug is difficult enough to produce—so once a shortage occurs it can be deeper than under AWP.
  > - **Takeaway:** "resistance to occurrence" does not guarantee "small magnitude"—a warning for policymakers.

- **Figure 7 (driving forces):** decomposes shortage-magnitude change into between-drug (a), between-GPO (b), and free-ride (n) forces; free-ride is proshortage while the other two are antishortage, compared across 8 scenarios.
- **Figure 8 + Table 2 (drug heterogeneity):** drugs with larger n are more shortage-prone. Category examples: large-n = alimentary-tract/metabolism/anti-infective drugs; small-n = orphan & newly-approved generics; large-a = most injectable generics; small-a = hormonal preparations, immunomodulators; large-b = sterile injectables, chemo agents, anesthetics; small-b = oral & topical generics.

6. Main Conclusions & Managerial Insights

> **Core conclusion:** ASP possesses **resistance** to shortages, especially for drugs with **few GPOs (small n, weak free-riding)** and **high supply difficulty (large a, b, strong aligning effect)**. Because the drugs that currently experience shortages tend to have exactly these traits, **ASP is unlikely to be the major cause**—contrary to the popular narrative that blames ASP.
> **Caveat:** for drugs with moderate supply difficulty, shortage magnitude under ASP can be larger when shortages do occur.

Policy implications:

- **Raising θ** (e.g., the 2019 proposal to lift the biosimilar markup from 6% to 8%) mitigates shortages but with diminishing returns, and must be weighed against higher Medicare spending; it should be **drug-specific**: drugs already resistant (ABn/ABN types) need little increase, while weakly resistant drugs (abn/abN) benefit more.
- ASP as a "market-data-driven" pricing scheme is highly relevant to CMS's 2026 "maximum fair price" negotiations.
- **GPO price consciousness α must be balanced:** over-emphasizing low wholesale prices hurts providers both long-term (shortages) and short-term (lower reimbursement income). Balance price against non-price factors—quality (QMM), service level (echoing Jia & Zhao 2017, Naumov et al. 2025).
- Provide shortage "alerts" for drugs with large **n** and low supply difficulty (weak resistance), e.g., anti-infective drugs, hormonal/immunomodulating agents.
- **Price ceilings / cost floors:** overly restrictive ceilings can trigger shortages; cost floors matter more for high-profit drugs; both effects are drug-specific.

7. Storyline

Overall a **progressive** structure with one pivotal **counterintuitive twist**:

- **Introduction:** real-world shortage pain point → the pro/con debate on ASP → a triple gap (no wholesale-pricing model, GPOs overlooked, scarce data) → the two research objectives.
- **Literature review (three parallel blocks):** drug shortages / reimbursement policy / price regulation. Distinguishes from Yurukoglu et al. (2017), which omits GPOs and the two price tiers; the "two-tier price structure" is the novelty.
- **Model (built backward, downstream → upstream):** a parsimonious model whose core innovations are the GPO objective (weighted members' profit + own profit, via price consciousness **α**) and the supply-allocation form with supply difficulty **a**, **b**.
- **Analysis (progression + twist):** equilibrium (Thm 1) → shortage measures → **unique dynamics (Thm 2)** → policy parameters (Thm 3). The "anomaly" appears here—supply difficulty should worsen shortages yet reduces them under ASP—and is resolved by decomposing the **free-ride vs. aligning** effects, the analytical hinge of the paper.
- **Numerical (validation + support):** estimate $n \approx 3$ (showing free-riding is weak in reality) → shortage regions → policy-parameter effects → magnitude caveat → force decomposition → drug heterogeneity. The numerics serve the against-the-grain claim that ASP is not the culprit.

- **Conclusion:** ASP resists shortages / is not the major cause; drug-specific policy advice; extensions.

8. Critique & Extension Ideas

Strengths (done well)

- **Solid modeling novelty:** captures the two-tier structure where GPOs set wholesale prices and the reimbursement price is a markup on the *average* wholesale price—distinct from regulated markets (energy, electricity) that regulate wholesale prices directly. A genuine increment to the economics/OM literature.
- **Fills a real gap:** brings GPOs—the least-understood, currently FTC/HHS-investigated party—into the model.
- **The free-ride / aligning decomposition is elegant and intuitive;** the counterintuitive main result (ASP resists shortages) challenges conventional wisdom.
- **Impressive data work:** overcomes pharmaceutical data opacity with ML to estimate n; strong policy relevance (2026 CMS negotiations, 2019 biosimilar amendment).

Weaknesses (open to question)

- **Manufacturers are not strategic (biggest limitation):** the supply equation (3) is a reduced-form approximation—supply is written as a function of prices plus a random factor, not derived from manufacturers' profit-maximizing game. Yet shortages fundamentally originate from manufacturers' exit / under-investment decisions; treating the supply side as a non-player is a notable shortcut.
- **Single-period & static:** collapsing the 2–3-year contract into one period cannot capture ASP's "reimburse on the prior two quarters' average" **dynamic lagged feedback**—precisely the mechanism behind the "price death spiral" (falling reimbursement → exits → deeper shortage).
- **Deterministic, price-independent demand:** reasonable for medically necessary drugs but limits generality; no demand-side dynamics.
- **Symmetric-GPO assumption:** though the authors argue asymmetry would strengthen results (Appendix K), heterogeneity in GPO bargaining power with manufacturers is not explicitly modeled.
- **A key dimension lacks empirical grounding:** a, b cannot be estimated from data, so supply-difficulty results rest on assumed ranges; no causal econometric test of the model's predictions.
- **The magnitude caveat (Fig. 6)** somewhat undercuts the "resistance" narrative but is discussed only briefly.

Extensions for a new paper (what to build next)

- **Endogenize manufacturer decisions:** model entry/exit, capacity investment, and quality as a strategic game so that shortage emerges from equilibrium supply—directly addressing the biggest gap; potential for a fuller game-theoretic paper.

- **Dynamic/multi-period model:** explicitly model ASP's two-quarter-lagged price–reimbursement feedback loop and analyze whether a "price spiral / death spiral" arises and its stability—a valuable theory extension.
- **Empirical validation:** as data transparency improves, test the model's predictions econometrically (e.g., n vs. shortage propensity; the causal effect of θ changes)—moving from "numerical calibration" to a genuine empirical paper.
- **Asymmetric GPOs and explicit bargaining:** model GPO–manufacturer bargaining, inter-GPO competition, and scale heterogeneity.
- **Endogenize non-price factors:** incorporate quality (QMM ratings) and service level (failure-to-supply clauses) into GPOs' procurement decisions, connecting to Naumov et al. (2025).
- **Price-ceiling / cost-floor policy experiments:** model CMS's 2026 "maximum fair price" as an explicit price ceiling and analyze its differential shortage impact across drug categories.

Prepared from a close reading of Zhao, Jia & Zhao (2025), *Reimbursement Policy and Drug Shortages*, Management Science (21 pp.), in preparation for a literature review.